Question 5: Bokeh Effect Implementation

Pushpendra Uikey

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Overview

This section covers the implementation of a simulated Bokeh effect. The goal is to artificially blur the background of an image while keeping a manually defined foreground object perfectly sharp. The algorithm applies a uniform, disc-shaped filter (of diameter 50 and 100 pixels) exclusively to the background regions.

Methodology

The primary challenge of this assignment is handling the blur at the boundaries—both at the edges of the image and at the border where the sharp foreground meets the blurred background. If we just blindly blur the whole image and paste the sharp foreground back on top, the colors from the foreground will illegally bleed out into the background blur.

To solve this efficiently, the implementation is split into a "fast path" and a "slow path" using a distance transform:

1. **Manual Masking:** An interactive Matplotlib script ('generate_masks.py') was used to hand-draw polygon masks around the foreground objects.

2. **Global Fast Blur:** The entire image is convolved with the circular disc kernel using a fast Fourier transform ('scipy.signal.fftconvolve'). This calculates the blur incredibly fast, but the pixels near the foreground boundary are "contaminated" by foreground colors.

3. **Distance Transform:** A distance map is calculated from the foreground mask. This tells us exactly how far away every background pixel is from the foreground.

4. **The Safe Zone (Fast Path):** Any background pixel that is further away from the foreground than the radius of the blur kernel is considered "safe". We simply copy the result directly from the fast global blur.

5. **The Unsafe Zone (Slow Path):** For the narrow band of background pixels close to the foreground (or the image edges), we must drop into a manual loop. For each pixel, the circular kernel is cropped so it never overlaps the foreground. The cropped kernel is then re-normalized (rescaled) so its weights sum to 1.0, ensuring the background doesn't darken at the edges.

Results

The algorithm was tested on three images: 'deep.jpg', 'lotus.jpg', and 'marigold.jpg'. In all cases, the foreground objects remain perfectly crisp while the background takes on a smooth, circular blur that increases in intensity from diameter 50 to diameter 100.

Deep (Candles)

The candles in the foreground remain sharp. Because of the boundary cropping logic, the bright light from the foreground candles does not bleed outwards into the dark background, keeping the edges clean. Notice how the distant candles in the bowl melt into soft, circular points of light, mimicking a real camera lens.

Figure 1: Bokeh effect applied to ‘deep.jpg‘. The background candles blur into circular discs.

Lotus

The lotus flower is cleanly isolated from the pond. The boundary logic successfully prevents the bright pink of the petals from smearing into the green water, even at the extreme $d = 100$ blur level.

Figure 2: Bokeh effect applied to ‘lotus.jpg‘. The foreground flower remains perfectly sharp.

Marigold

The intricate edges of the marigold are preserved. The background leaves are aggressively smoothed out, effectively drawing the viewer’s eye directly to the center flower.

Figure 3: Bokeh effect applied to ‘marigold.jpg‘.